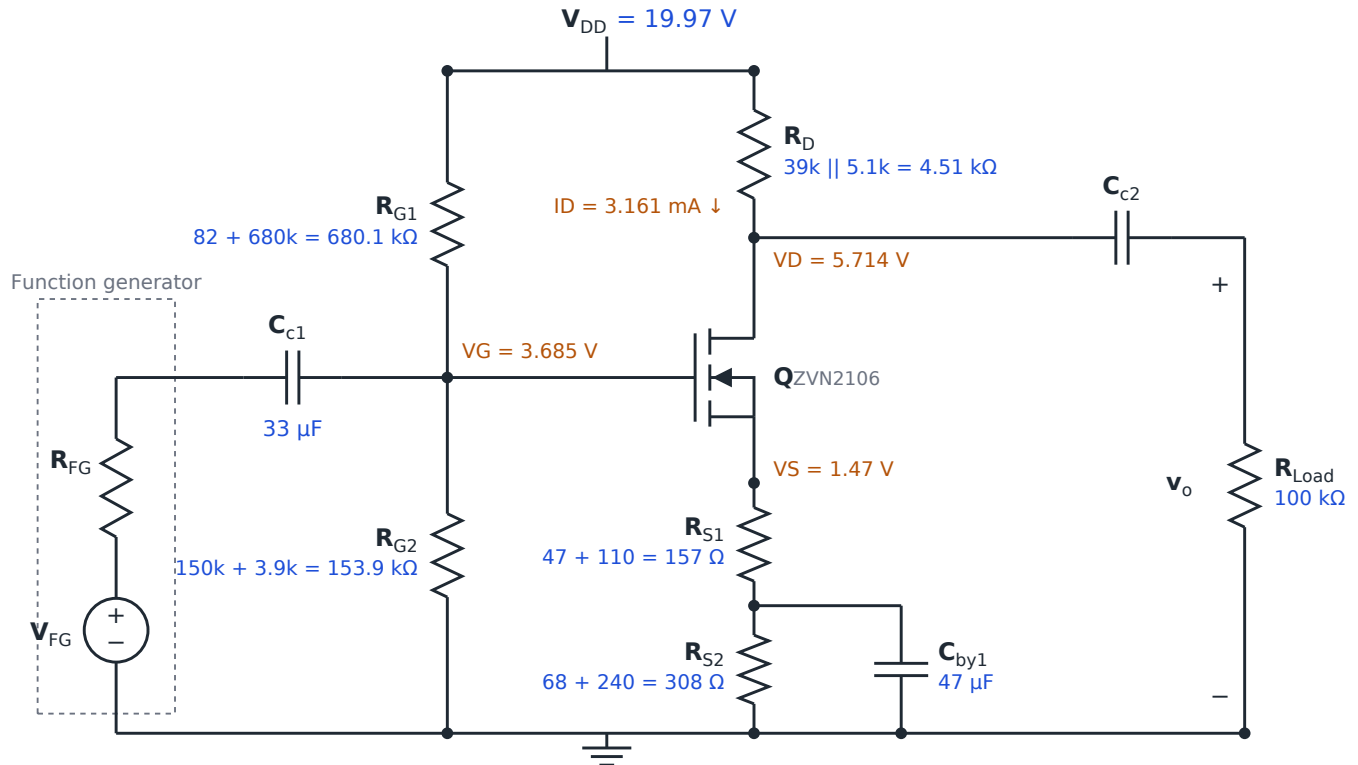


Common-Source Stage — Optimized Design

Nominal score 0.9040

ECE321 Lab 3 · generated 2026-10-07 23:04

Monte Carlo mean 0.8910 · pass rate 100 %



■ component values (from Circuit_Sol)

■ DC operating point (from analyzeCSAmpModel)

■ Cc2 value not part of model

Analyzed values

Quantity	Nominal	MC min	MC max
Overdrive voltage V_{OV}	275.3 mV	231.5 mV	312.5 mV
Drain current I_D	3.161 mA	2.419 mA	3.89 mA
Gate voltage V_G	3.685 V	3.415 V	3.966 V
Source voltage V_S	1.47 V	1.133 V	1.803 V
Drain voltage V_D	5.714 V	2.503 V	9.107 V
Drain-source V_{DS}	4.244 V	699.4 mV	7.959 V
Gate-source V_{GS}	2.215 V	1.983 V	2.424 V
Transconductance g_m	22.96 mS	20.07 mS	25.5 mS
Input resistance R_i	125.5 k Ω	119.1 k Ω	132.2 k Ω
Open-circuit gain A_{v_o}	-22.49 V/V	-24.08 V/V	-20.92 V/V
Output resistance R_o	4.51 k Ω	4.28 k Ω	4.68 k Ω
Loaded gain A_v	-21.51 V/V	-22.99 V/V	-20.04 V/V
Power R_D	39.86 mW	23.14 mW	59.83 mW
Power R_{G1}	390 μ W	346.8 μ W	438.3 μ W
Power R_{G2}	86.02 μ W	69.98 μ W	102.1 μ W
Power R_{S1}	1.099 mW	595.7 μ W	1.585 mW
Power R_{S2}	2.398 mW	1.461 mW	3.735 mW
Swing margin, downward	2.369 V	-1.206 V	6.128 V
Swing margin, upward	12.04 V	8.781 V	15.12 V
Distortion estimate HD_2	1.47 %	1.2 %	1.93 %
Threshold voltage V_t	2 V	1.717 V	2.234 V

Score breakdown (nominal)

Category	Score	Weight
ID in range	0.9736	1
RI in range	0.9980	1
RO in range	0.9116	1
Av in range	0.9868	1
In saturation	0.9108	1
Saturation margin	1.0000	1
RG1 Power	0.2499	0.25
RG2 Power	0.2500	0.25
RS1 Power	0.2499	0.25
RS2 Power	0.2497	0.25
RD Power	0.2447	0.25
Low Distortion	0.4334	1
Meets req	1.0000	1

Total (nominal) 0.9040

Monte Carlo

Analysis samples	200 (invalid: 0)
Meets req pass rate	100.0 %
scoreCSAmpModel mean	0.8910
scoreCSAmpModel min / max	0.8764 / 0.8973
Deviation σ (parts / FET)	2 % / 5 %
Reps per score	100